Dowon Kim

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Education

B.S. in Global School of Media | GPA: 4.3/4.5 Mar 2020 - Feb 2026
Soongsil University

Research Experience

LG AI Research Jan 2026 - present
Robotics Data Engineer Intern, Physical Intelligence Lab

Soongsil University Jun 2024 - Dec 2025
Undergraduate Student Researcher, Reality Lab

Awards

1st Place at ARNOLD Challenge, CVPR2025 Embodied AI Workshop Jul 2025
Grand Prize, 2025 Gyeonggi Youth Gap Year Program Nov 2025
Departmental Honors, Soongsil Univ Feb 2026

Publications

[D2] **A Deep Learning-Based Web Application for Automatic Detection and Diagnosis of Skin Diseases**, Sebin Lee, Dowon Kim, Geunchan Park, Dohun Kim, Iksu Kim
Journal of The Korea Contents Association 2025

[D1] **Quantitative Analysis of Multi-Viewpoint Configuration Effects in Robotic Manipulation Policy Learning**, Dowon Kim, Sangmin Lee, Sungyong Park, Heewon Kim
Journal of Broadcast Engineering 2025

Projects

Vision-Language-Action Policy Learning on ARNOLD Benchmark Apr 2025 - May 2025
Fine-tuned Vision-Language-Action models for embodied manipulation on the ARNOLD benchmark. Improved robustness and generalization through targeted data augmentation, outperforming the PerAct baseline in continuous control tasks within Isaac Sim.

Large-Scale Sim-to-Real Robotics Data Pipeline Jan 2026 - Present
Co-designing and building an end-to-end R2S2R pipeline using NVIDIA Isaac Sim at LG AI Research. Developing the full data loop from real-world collection to synthetic augmentation and deployment validation, focusing on scalable architecture and sim-to-real robustness.

Visual Guidance for Failure Modes in VLA Models Oct 2025 - Present
Analyzing systematic failure modes of VLA models and proposing a visual guidance framework to enhance agent reasoning and control stability across long-horizon manipulation tasks.

Humanoid Mobile Manipulation System Design Aug 2025 - Dec 2025
Designed and implemented a humanoid mobile manipulation system to study the full robotics lifecycle, from sim-to-real policy deployment to large-scale dataset generation and evaluation.

One-Step Diffusion-based Anomaly Detection Mar 2024 - Jun 2024
Developed a one-step denoising variant of DiffusionAD for real-time structural anomaly detection in biological products, enabling efficient inspection of irregular industrial defects.

IoT-based Smart Parking Management System Sep 2024 - Dec 2024
Built a Raspberry Pi–based smart parking system integrating license plate recognition, GPIO-controlled hardware modules, and a Flask-based management interface for real-time monitoring and control.

Scholarship

National Scholarship for STEM (Full Scholarship), KOSAF 2020 - 2026

Media

Soongsil Univ. 1st Place at CVPR '25 ARNOLD Challenge, The JoongAng Jul 2025

Skills

Languages: Python, C++, C, Linux Shell Script
Tools: Isaac Sim, ROS2, Ubuntu
Robot: Franka, RX1, AI worker, OMY